

# Cellulose extraction and quantification from eucalyptus leaves in the laboratory

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## Method Article

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# Abstract

This protocol aims to extract cellulose from plant material using a nitric acid digestion and quantification using spectrophotometry. This protocol has been modified from existing protocols so that extraction of cellulose can be performed in 2 mL microcentrifuge tubes using ground plant material as the starting product. Key stages of the procedure include the nitric acid digestion and removal of breakdown products where approximately 100 samples can be digested in a day. The procedure is concluded with a conversion of cellulose to glucose, and finally quantification using spectrophotometry where approximately 200 samples can be quantified in a day.

## Introduction

Cellulose is a polymer chain made of glucose units that provides rigidity to the cell wall (Somerville, 2006). The conversion of cellulose to glucose in this experiment, as outlined by Kumar & Turner (2015), is what enables the quantification of cellulose. Cellulose plays a role in the sustainability and combustibility components of eucalypt leaf flammability (Ciccioli et al., 2014; Lowden & Hull, 2013). The former refers to the duration of flaming combustion of a leaf, and the latter refers to the amount of energy released during combustion (Martin et al., 1993). We have drawn on a range of existing protocols to optimize a cellulose quantification protocol that uses ground plant material and is scaled down to digestion in a 2 mL microcentrifuge tube while allowing a high throughput of samples. This small scale ensures that lower amounts of concentrated acid are used, preventing excessive waste generation and lower risk to the users of this protocol. As this protocol uses strong acids, it requires sufficient and adequate PPE to be completed. A safe work procedure and risk assessment should be completed before this experiment is executed. Additionally, pipette tips with barriers are required when working with strong acids and acetone to prevent damage to the internal mechanisms of auto-pipettes.

## Reagents

- Ethanol HPLC grade (99%; v/v) CAS-No: 64-17-5
- Acetic acid HPLC grade (80%; v/v) CAS-No: 64-19-7
- Nitric acid HPLC grade (69%; v/v) CAS-No: 7697-37-2
- Acetone HPLC grade CAS-No: 67-64-1
- Sigma-Aldrich Anthrone CAS-No: 90-44-8
- AnalaR Glucose CAS-No: 50-99-7
- Sulphuric acid HPLC grade (97%; v/v) CAS-No: 7664-93-9

## Equipment

- Leaf samples
- 20 ml polyethylene vials for grinding samples
- Metal beads for grinding samples
- Cryogenic bead grinder (or other method for grinding leaves)
- 2 mL round-bottom microcentrifuge tubes
- 15 mL screw capped tubes or similar
- Precision balance (to four decimal places)
- Heating block that fits 2 mL round bottom tubes
- Eppendorf Centrifuge 5804 or similar
- Parafilm Laboratory Film or similar
- Auto-pipettes and barrier pipette tips
- Glass stirring rod
- Boiling water bath
- Ice
- Spectrophotometer e.g. SPECTROstar<sup>Nano</sup> BMG Labtech with software version 5.70 R2 or similar
- 96-well microplates

## Procedure

This method is based on Brendel et al. (2000), Kumar & Turner (2015), and Greer et al. (2018) to develop a simple and rapid procedure for cellulose quantification using ground plant material.

1. Places a few leaves in 20 mL polyethylene vial with 4-5 metal beads for grinding and cap appropriately. Grind leaves using a cryogenic bead grinder.
2. Take 2 mL round-bottomed microcentrifuge tube and using a pin or sharp tweezers make two holes in the lid.
3. Weigh out ~20 mg of ground plant material into a 2 mL round-bottomed microcentrifuge tube.

4. Add 400  $\mu\text{L}$  of acetic acid (80%; v/v, if using glacial acetic acid prep to this concentration beforehand) and 40  $\mu\text{L}$  of concentrated nitric acid (69%; v/v) to the sample and leave with lids open for 15 minutes.
5. Cap tube tightly and move to heating block, start with heating block at room temperature and set temperature to 120 C. NOTE: it is important to start with the heating block at room temperature to prevent overpressure inside the microcentrifuge tubes and potential spillage of the acid.
6. Once heating block reaches 120 C boil samples for 30 minutes tapping the top of each tube every 5 minutes to prevent condensation building up on the lid.
7. Move tubes to rack to cool.
8. Wash the sample as follows:

Add the required reagent and invert to mix using parafilm to cover holes in lid. Centrifuge for 5 minutes at 7000 rpm and discard supernatant.

  - a. 1 x 1000  $\mu\text{L}$  ethanol, to remove extraction breakdown products,
  - b. 1 x 1000  $\mu\text{L}$  deionized water, to remove traces of nitric acid,
  - c. 1 x 1000  $\mu\text{L}$  ethanol,
  - d. 1 x 1000  $\mu\text{L}$  acetone.
9. Cover rack with samples with aluminium foil and leave to dry in fume hood overnight. Figure 1 displays some examples of cellulose after the digested and washed products have dried.
10. Make a 67% sulphuric acid solution by slowly adding 335 mL concentrated sulphuric acid to 165 mL milli-Q water and allow solution to cool. NOTE: Always add acid to water and never the other way around.
11. Add 1 mL of the 67% sulphuric acid solution to the dried cellulose, only add acid to the number of samples you plan to process that day.
12. Shake at room temperature for 1 hour until no material can be seen.
13. Prepare the glucose standards as outlined in Kumar & Turner (2015). These standards should have concentrations of 0, 10, 20, 40, 60, 80 and 100  $\mu\text{g/mL}$  of glucose.
14. In a new 15 mL screw capped tube add 500  $\mu\text{L}$  of milli-Q water and add 20  $\mu\text{L}$  of a sample to the water. Additionally, aliquot 500  $\mu\text{L}$  of each glucose standard into a 2 mL screw capped tube.
15. Prepare 100 mL of a 0.3% anthrone in concentrated sulphuric acid solution over ice (stability ~ 3 hours). Mix anthrone into sulphuric acid using a glass stirring rod.

16. Add 1 mL of the anthrone solution to each test sample and standard slowly down the side of the microcentrifuge tube to prevent mixing of samples as much as possible. Cap tightly and mix via inversion. NOTE: tubes will get very hot, wear appropriate PPE.
17. Place in boiling water bath for five minutes and cool on ice. CRITICAL: It is important not to exceed the five minute timeframe when boiling the samples as they will lose their blue/green colouring and go an opaque white which is no longer viable for measurement.
18. Place 200 µL of sample in microplate and record OD620.
19. Generate a regression curve using the glucose standard readings and determine glucose concentration of each sample.
20. Use equation 1 to determine the concentration of cellulose in the ground plant material.

Equation 1: Cellulose (% of ground plant material) =  $\text{glucose } (\mu\text{g/mL}) / (\text{weight of samples } (\mu\text{g}) \times 100 \times 50 \text{ (to get total amount in 1 mL)})$

## Troubleshooting

See Figure 2 for troubleshooting advice for this protocol.

## Time Taken

Nitric Acid Digestion – Total Time 2 hours for 48 samples

- Weigh out samples into 2 mL microcentrifuge tubes, approximately 1 hour
- Add acids and digest samples, approximately 1 hour

Washing Cellulose – Total Time 2 hours for 48 samples

- Wash the cellulose to remove breakdown products using four wash steps, each step requires the addition of a reagent, vortexing, centrifuging and removal of supernatant.

Quantification of Glucose – Total Time 5.5 hours for 200 samples. This is more easily achieved with two people.

- Add sulphuric acid to dried cellulose and prepare glucose standards and 15 mL tubes for samples to be transferred to, 1.5 hours
- Add sample to water, 1 hour
- Prepare anthrone solution and add to each of the 15 mL tubes, 1 hour
- Boil samples in water for 5 minutes and transfer to ice, 1 hour
- Aliquot samples into 96 well plates and quantify glucose, 1 hour

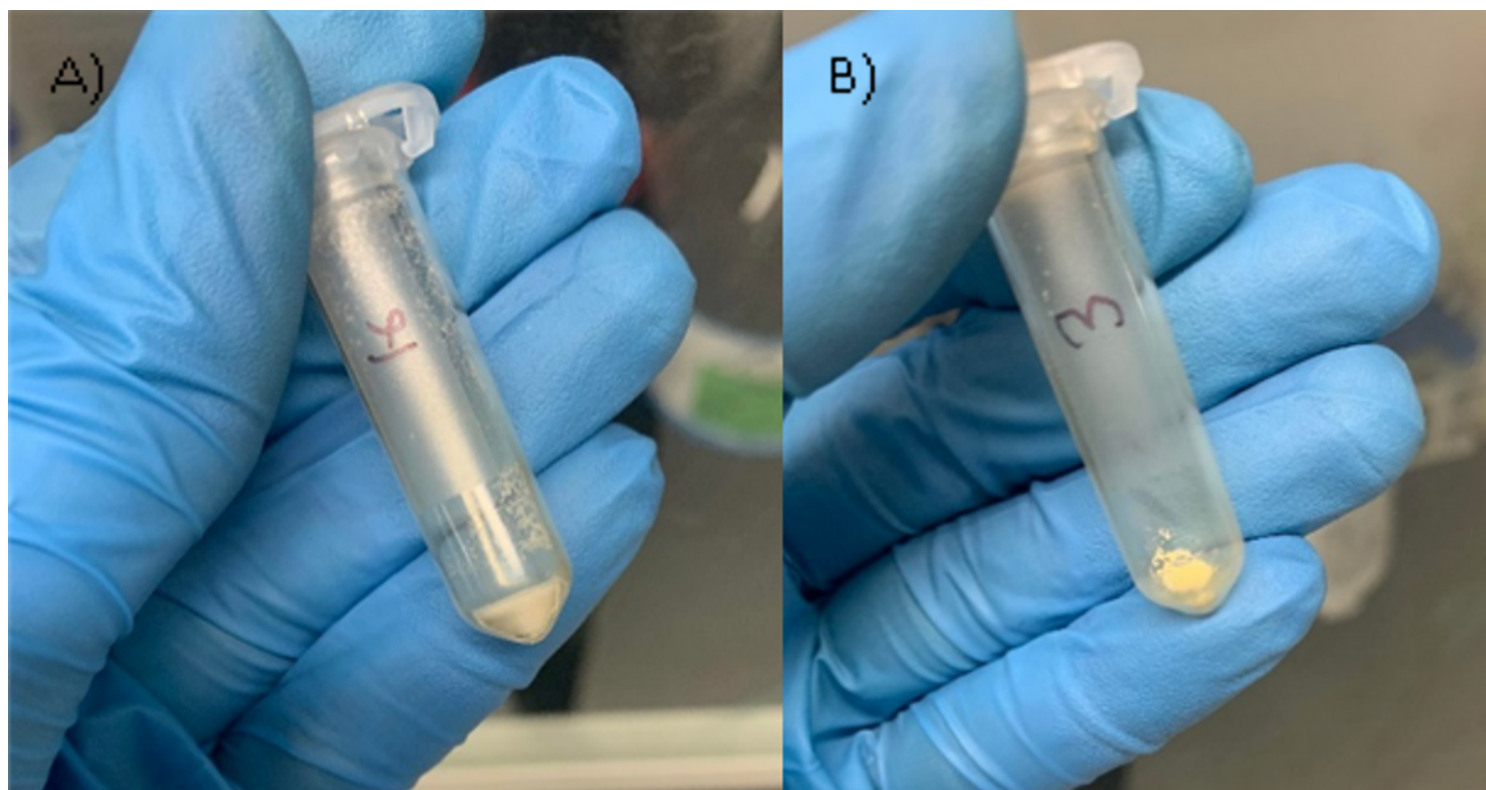
## Anticipated Results

Samples should range from a pale yellow to dark blue in colour after boiling for 5 minutes (Figure 3). Final cellulose content as a percentage of the starting dried plant material ranged from 0.07-74%.

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## Figures



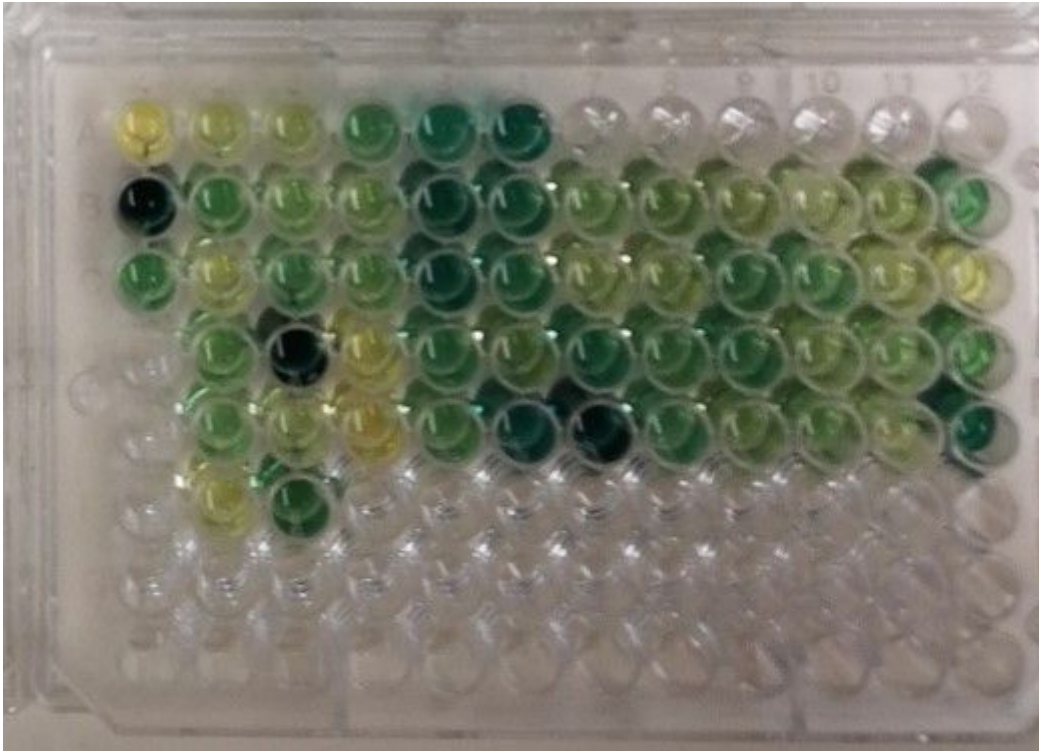
**Figure 1**

Examples of cellulose after digestion and washing. A) pure cellulose, B) plant material.

| Step | Common Issues                                             | Potential Reasons                                                                                                   | Solutions                                                                                                                                                                                                                                                                                                 |
|------|-----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5/6  | 2 mL tubes pop open during digestion                      | Pressure in the vessel has built up too quickly                                                                     | It is important that there are holes in the lids of the Eppendorf tubes and that the time is taken to allow the tubes to heat up from room temperature.                                                                                                                                                   |
| 6    | Samples have dried out during digestion                   | Acid has evaporated out of the holes in the lid and the sample has dried out meaning there was incomplete digestion | Add additional nitric acid and acetic acid and continue to digest for an additional 30 minutes. The acid mixture should be a yellow colour at the end of the digestion and the sample should be white. Adding additional acid has not proved problematic as all acid is removed during the washing phase. |
| 19   | Sample turns opaque white instead of blue/green in colour | Boiled for too long                                                                                                 | Sample is no longer viable and will have to be re-extracted from a new sample of plant material. Use a foam rack to hold samples in when submerged in boiling water so all samples can be removed rapidly and easily from boiling water.                                                                  |

**Figure 2**

Common issues and troubleshooting advice for this protocol.



**Figure 3**

Examples of cellulose after digestion and washing. A) pure cellulose, B) plant material.